

1 Testbed construction
104-gene HOTAIR-interactome panel
30 candidate + 44 non-candidate
+ 30 T2DM control

2 Dual-source S-PrediXcan TWAS
GTEx v8 MASHR (Nerve_Tibial, Whole_Blood)
vs eQTLGen whole blood (N = 31,684)

3 Cross-source agreement
Spearman rho on Z-scores
pairwise direction consistency

4 Two-axis decomposition
resource / sample-size axis
tissue-context axis

5 Calibration & replication
30 disease-agnostic housekeeping controls
FinnGen R13 vs UK Biobank (DR)

6 Evidence integration
STREGA-based TWAS reporting checklist
(Additional file 1: Table S7)